How Field of View Affects Awareness of an Avatar During a Musical Task in Augmented Reality

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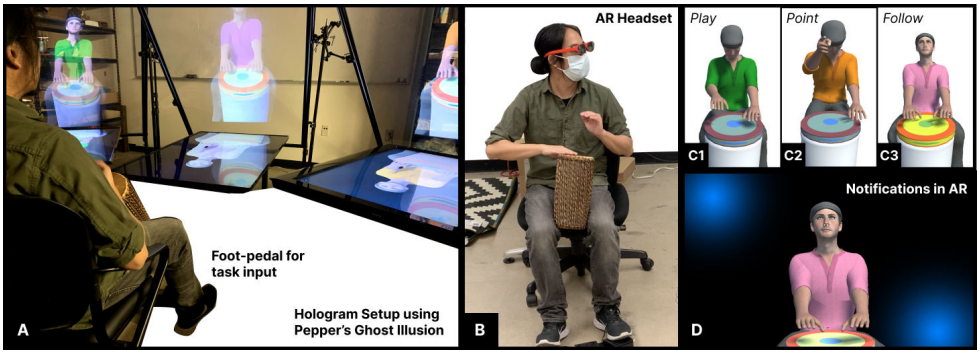


Figure 1: (A) Hologram drum circle, (B) AR glasses drum circle setup, (C) AR with notification, and (D) AR without notification

Abstract

To investigate how field of view (FOV) affects how players notice the communicative gestures of their partner's avatar in a musical task, we conducted an experiment that compares several AR technologies with varying FOV. We measured response time to communicative gestures, co-presence score, and task enjoyment in three different AR scenarios: holograms, AR headset, and an AR headset with notifications of the avatar's intention to gesture. Results suggest that the hologram setup had the fastest response time and highest ratings for sense of co-presence and task enjoyment. Notification tasks slowed response time, but noticeably improved co-presence with the avatar.

Index Terms: Human-centered computing—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Interaction paradigms—Collaborative Interaction; Applied computing—Sound and music computing

1 Introduction and Related Works

Recently there has been a significant uptake in the individual use of AR headsets. This has resulted in increased interest in studying the interactions between humans and virtual content in AR. While there have been challenges with AR technology, such as a limited FOV, gaining knowledge of how humans understand and interact with virtual content given these constraints is important as the technology continues to improve. Related research in AR investigates how the user interprets information which is outside of the FOV [5, 7] and much of the research focuses on improving task performance by using message notifications [1–3, 6]. Musical collaboration, being highly reliant on communicative gestures, is one of the scenarios that researchers use to investigate interaction with avatars [4]. In a real-life musical collaboration activity, a drum circle for example, participants need to listen to the music that participants is playing and they need to be aware of gestural interaction within the group including facial expression, hand gestures, and eye gaze. These cues are easy to notice in face-to face situations, however, it is difficult to notice social cues in a multi-avatar collaboration situation when avatars are outside of the FOV. Nevertheless, communication is a crucial component of maintaining the social dynamic in a musical collaboration experience. To investigate this further we ask three research questions between three conditions Hologram (no limit of FOV), AR, AR Notifications: RQ1: How fast can a player notice an avatar drummer when the avatar is following the player (Avatar is looking at player)? RQ2: What is the Co-Presence score (by Aggregated Social Presence Scores questionnaire [5]) in avatars? RQ3: Does the limited FOV affect task enjoyment?

2 Method

To answer our research questions we conducted an experiment to compare the musical experience and social presence with avatars in different FOV situations in AR. We conducted a 2x3 within-subjects experiment to simulate drum circle activities with three avatars. We recruited a total of 12 participants for our experiment (6 male, 6 female). We used two hardware set ups. The first was a Hologram Drum Circle (HDC) (Figure 1,it is an pepper's ghost illusion [1] hardware set up which is A) run with the laptop PC equipped with an Intel 10700k CPU, an Nvidia 1070 GPU, and 32GB of memory with three televisions. The second utilized NReal AR glasses Drum Circle (ARDC) (Figure 1, B) [2]. In the ARDC set up, we placed three virtual avatars in the AR environment. One is in the middle, and the other two avatars are on both sides beyond 52 degrees (outside

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[1] https://www.instructables.com/The-Peppers-Ghost-Illusion/
[2] https://www.nreal.ai/light/

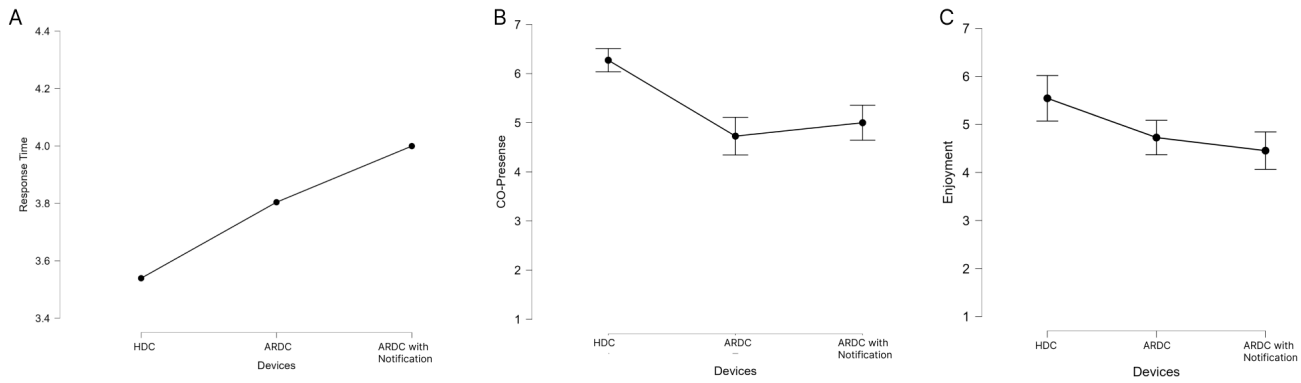


Figure 2: (A) Response time (B) Co-presence Score (C) Task enjoyment

FOV). In this setup, we have two contexts: 1) The Context 1 is with visual notification which uses blue hints to indicate when an out of FOV avatar is gesturing (upper corner means following, down corner means pointing at you (Figure 1 D), 2) and the Context 2 has no visual notification. In the HDC setup, we put three holograms in front of a participant with the same FOV as ARDC. HDC is to simulate playing the drum circle with avatars but without an FOV limitation. Avatar models and animations are from the Mixamo character and animation platform[3]. During the experiment, participants needed to answer whether they noticed the AR avatars' gestures during a drum circle activity in both setups and contexts by hitting a foot pedal placed in front of them. Participants played a hand drum while making their selections during the entirety of the task to simulate the cognitive load present in real drum circle scenarios. Based on observed drum circles avatars had three kinds of animated gestures. The first gesture was pointing at participants but looking at their own drum. The second was to look at their drum only while also playing it. Finally, the third was to look at the participants and nod their heads, indicating a following gesture that participants needed to select(Figure 1 C), 2). Every drum circle session had 10 rounds of tasks in the experiment.

3 Results and Discussion

We measured the response time when the participants noticed the following avatar. We collected 10 rounds of data from each participant and measured the mean response time after all rounds. To analyze the data, a repeated measures Analysis of Variance (RM-ANOVA) was conducted on the data. While there were no significant differences, initial results seem to indicate that in the HDC task, people were faster to notice the following avatar. The mean response time in the ARDC task with the notification showed the slowest response time (Figure 2, A). We interpret this result to be due to the disturbance imparted by the notification in our system. After the participants finished each task, we gave them a social aggregate and task enjoyment questionnaire. Quantitative scores between 1 and 7 (1 - lowest, 4 - neutral, and 7 - highest) were indicated for each measure of co-presence and enjoyment. The results of the co-presence score between the three devices showed that HDC had the highest score of the three (m= 6.175) and ARDC without notifications showed the lowest score (m= 4.8). HDC had a significantly higher co-presence score than the other tasks (a RM-ANOVA revealed a significant difference among the score means $F(2,30)=6.175$ $p<0.05$, $\eta_p^2=0.292$) (Figure 2 B). Results suggest the limitation of FOV does affect co-presence with avatars, and notifications may help to improve co-presence while using AR devices. The task enjoyment score also indicated that the HDC had the highest score (m= 5.8), and the ARDC with notifications had the lowest score (m= 4.5) (Figure 2 C).

In summary, results indicate that HDC had the highest co-presence score, lowest response time, and highest task enjoyment score. We interpret that the limitation of FOV does negatively affect the musical collaboration with the avatars. Also, the results of the ARDC task suggest that co-presence can be improved with the use of notifications, perhaps at the cost of enjoyment.

Our goal is to investigate how FOV affects how players notice the communicative gestures of their partner's avatar in a musical task. Furthermore, we want to improve the sense of co-presence with the avatars and the player's task enjoyment in limited FOV situations. According to this research result, there is a potential improvement in the AR with the notification system; therefore, we will improve the AR with the notification system in the future. Nevertheless, in future experiments, we will increase our sample size. To help us get more accurate and detailed results.

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